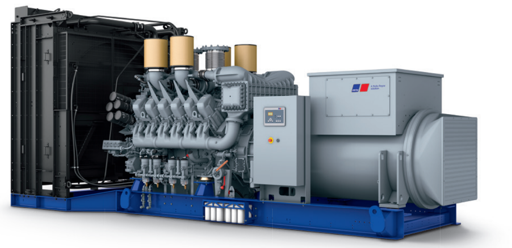
Diesel Generator Set

mtu 12V4000 DS2000

380V – 11 kV/50 Hz/grid stability power/ fuel consumption optimized/12V4000G24F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits
- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support
- Global product support offered

Standards
- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating
- System ratings: 1820 kVA - 1880 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)
- Engine-generator set tested to ISO 8528-5 for transient response
- 100% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available
- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions
- Fuel consumption optimized

Certifications
- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary

A Rolls-Royce solution

Application data[1)]

Engine

Manufacturer	*mtu*
Model	12V4000G24F
Type	4-cycle
Arrangement	12V
Displacement: l	57.2
Bore: mm	170
Stroke: mm	210
Compression ratio	16.4
Rated speed: rpm	1500
Engine governor	ECU 9
Max power: kWm	1575
Air cleaner	dry

Fuel system

Maximum fuel lift: m	5
Total fuel flow: l/min	16

Fuel consumption[2)]	l/hr	g/kwh
At 100% of power rating:	364.3	192
At 75% of power rating:	274.7	193
At 50% of power rating:	189.8	200

Liquid capacity (lubrication)

Total oil system capacity: l	260
Engine jacket water capacity: l	160
Intercooler coolant capacity: l	40

Combustion air requirements

Combustion air volume: m^3/s	1.8
Max. air intake restriction: mbar	50

Cooling/radiator system

Coolant flow rate (HT circuit): m^3/hr	56
Coolant flow rate (LT circuit): m^3/hr	30
Heat rejection to coolant: kW	580
Heat radiated to charge air cooling: kW	260
Heat radiated to ambient: kW	75
Fan power for electr. radiator (40°C): kW	38

Exhaust system

Exhaust gas temp. (after turbocharger): °C	440
Exhaust gas volume: m^3/s	4.5
Maximum allowable back pressure: mbar	85
Minimum allowable back pressure: mbar	30

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	Fuel consumption optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA52.3 S6 (Low voltage Leroy Somer standard)	380 V	1504	1880	2856	1472	1840	2796
	400 V	1504	1880	2714	1472	1840	2656
	415 V	1504	1880	2615	1472	1840	2560
Marathon 743RSL7091 (Low voltage Marathon)	380 V	1496	1870	2841	1456	1820	2765
	400 V	1504	1880	2714	1456	1820	2627
	415 V	1496	1870	2602	1456	1820	2532
Marathon 744RSL7092 (Low voltage Marathon oversized)	380 V	1496	1870	2841	1456	1820	2765
	400 V	1504	1880	2714	1456	1820	2627
	415 V	1496	1870	2602	1456	1820	2532
Marathon 1020FDH7096 (Medium volt. marathon)	11 kV	1496	1870	98	1456	1820	96
Leroy Somer LSA53.2 VL7 (Medium volt. Leroy Somer)	11 kV	1504	1880	99	1472	1840	97

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).

2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.